VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS301

Operating Systems & System Programming - Comprehensive Syllabus Lecture

Syllabus Focus: CS301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Operating Systems & System Programming. Inclu

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Kernel Architecture & Process Scheduling

" Syllabus Scope: Microkernel vs Monolithic, process states, PCB, context switching, CPU scheduling: FCFS

Unit 2: Concurrency & Synchronization Primitives

" Syllabus Scope: Critical section problem, Peterson algorithm, hardware test-and-set, semaphores, classi

CS301: Operating Systems & System Programming - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Deadlock Characterization & Prevention

" Necessary conditions, resource allocation graphs, Banker algorithm for safety and avoidance, detection and r

Unit 4: Memory Hierarchies & Virtual Memory

" Address translation, paging, segmentation, TLB, demand paging, page replacement FIFO, LRU, Optimal, thrashin

Unit 5: Storage Subsystems & File Layouts

" Inode structures, contiguous/linked/indexed file allocation, free-space bitmaps, disk scheduling SSTF, SCAN,

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the average turnaround and waiting times for Round Robin quantum=4ms on a given 5-process burst
2. Prove how the Banker Algorithm guarantees deadlock avoidance by verifying safe state conditions on an
3. Calculate the Effective Memory Access Time EMAT given TLB hit ratio 95%, TLB lookup 10ns, and main mem

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Abraham Silberschatz, Peter B. Galvin - Operating System Concepts Wiley 10th Ed
* Andrew S. Tanenbaum - Modern Operating Systems Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students